SQL TEST

1. List all customers ordered by customerName

Query :- select * from customers order by customerName;

Results :-

```
mysql> select * from customers order by customerName;
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
| customerNumber | customerName | contactLastName | contactFirstName | phone      | addressLine1 | addressLine2 | city   | state | postalCode | country | salesRepEmployeeNumber | creditLimit |
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
|           2001 | ABC Corp     | Lee             | Chris            | 1111111111 | Street 1     | NULL         | NYC    | NY    | 10001      | USA     |                   1002 |    50000.00 |
|           2003 | Global Inc   | Singh           | Raj              | 3333333333 | Street 3     | NULL         | Delhi  | NULL  | 110001     | India   |                   1002 |    60000.00 |
|           2002 | XYZ Ltd      | Kim             | Pat              | 2222222222 | Street 2     | NULL         | London | NULL  | NW1        | UK      |                   1003 |    70000.00 |
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
3 rows in set (0.00 sec)
```

2. Display all products ordered by buyPrice descending

Query :- select * from products order by buyPrice DESC;

Results :-

```
mysql> select * from products order by buyPrice DESC;
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
| productCode | productName  | productLine | productScale | productVendor | productDescription | quantityInStock | buyPrice | MSRP  |
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
| P101        | Toyota Camry | Cars        | 1:18         | Toyota        | Sedan model        |             100 |    20.00 | 30.00 |
| P102        | Honda Civic  | Cars        | 1:18         | Honda         | Compact model      |             150 |    18.00 | 28.00 |
| P201        | Yamaha Bike  | Bikes       | 1:12         | Yamaha        | Sport bike         |             200 |    10.00 | 15.00 |
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
3 rows in set (0.00 sec)
```

3. Show all employees ordered by lastName

Query :- select * from employees order by lastName;

Results :-

```
mysql> select * from employees order by lastName;
+----------------+----------+-----------+-----------+------------------+------------+-----------+-----------+
| employeeNumber | lastName | firstName | extension | email            | officeCode | reportsTo | jobTitle  |
+----------------+----------+-----------+-----------+------------------+------------+-----------+-----------+
|           1003 | Brown    | Mike      | x103      | mike@company.com | 2          |      1001 | Sales Rep |
|           1002 | Doe      | Jane      | x102      | jane@company.com | 1          |      1001 | Sales Rep |
|           1001 | Smith    | John      | x101      | john@company.com | 1          |      NULL | Manager   |
+----------------+----------+-----------+-----------+------------------+------------+-----------+-----------+
3 rows in set (0.05 sec)
```

4. List all offices sorted by country

Query :- select * from offices order by country;

Results :-

```
mysql> select * from offices order by country;
+------------+----------+------------+--------------+--------------+-------+---------+------------+-----------+
| officeCode | city     | phone      | addressLine1 | addressLine2 | state | country | postalCode | territory |
+------------+----------+------------+--------------+--------------+-------+---------+------------+-----------+
| 2          | London   | 9876543210 | Baker St     | NULL         | NULL  | UK      | NW1        | EMEA      |
| 1          | New York | 1234567890 | 5th Ave      | NULL         | NY    | USA     | 10001      | NA        |
+------------+----------+------------+--------------+--------------+-------+---------+------------+-----------+
2 rows in set (0.01 sec)
```

5. Get all orders sorted by orderDate descending

Query :- select * from orders order by orderDate desc;

Results :-

```
mysql> select * from orders order by orderDate desc;
+-------------+------------+--------------+-------------+---------+----------+----------------+
| orderNumber | orderDate  | requiredDate | shippedDate | status  | comments | customerNumber |
+-------------+------------+--------------+-------------+---------+----------+----------------+
|        3003 | 2024-03-01 | 2024-03-06   | NULL        | Pending | NULL     |           2003 |
|        3002 | 2024-02-01 | 2024-02-05   | 2024-02-04  | Shipped | NULL     |           2002 |
|        3001 | 2024-01-01 | 2024-01-05   | 2024-01-03  | Shipped | NULL     |           2001 |
+-------------+------------+--------------+-------------+---------+----------+----------------+
3 rows in set (0.02 sec)
```

6. Show customer names with their sales representative

Query :- select c.customername, e.firstname, e.lastname from customers c join employees e on c.salesrepemployeenumber = e.employeenumber;

Results :-

```
mysql> select c.customername, e.firstname, e.lastname from customers c join employees e on c.salesrepemployeenumber = e.employeenumber;
+--------------+-----------+----------+
| customername | firstname | lastname |
+--------------+-----------+----------+
| ABC Corp     | Jane      | Doe      |
| Global Inc   | Jane      | Doe      |
| XYZ Ltd      | Mike      | Brown    |
+--------------+-----------+----------+
3 rows in set (0.02 sec)
```

7. List all products with their product line

Query :- select productName, productLine from products;

Results :-

```
mysql> select productName, productLine from products;
+--------------+-------------+
| productName  | productLine |
+--------------+-------------+
| Toyota Camry | Cars        |
| Honda Civic  | Cars        |
| Yamaha Bike  | Bikes       |
+--------------+-------------+
3 rows in set (0.02 sec)
```

8. Display orders with customer names

Query :- select o.ordernumber, c.customername
from orders o
join customers c
on o.customernumber = c.customernumber;

Results :-

```
mysql> select o.ordernumber, c.customername
    -> from orders o
    -> join customers c
    -> on o.customernumber = c.customernumber;
+-------------+--------------+
| ordernumber | customername |
+-------------+--------------+
|        3001 | ABC Corp     |
|        3002 | XYZ Ltd      |
|        3003 | Global Inc   |
+-------------+--------------+
3 rows in set (0.00 sec)
```

9. Show order details with product names

Query :- select od.ordernumber, p.productname, od.quantityordered
from orderdetails od
join products p
on od.productcode = p.productcode;

Results :-

```
mysql> select od.ordernumber, p.productname, od.quantityordered
    -> from orderdetails od
    -> join products p
    -> on od.productcode = p.productcode;
+-------------+--------------+-----------------+
| ordernumber | productname  | quantityordered |
+-------------+--------------+-----------------+
|        3001 | Toyota Camry |              10 |
|        3001 | Yamaha Bike  |               5 |
|        3002 | Honda Civic  |               7 |
|        3003 | Toyota Camry |               3 |
+-------------+--------------+-----------------+
4 rows in set (0.02 sec)
```

10. List payments with customer names

Query :- select p.checknumber, c.customername, p.amount
from payments p
join customers c
on p.customernumber = c.customernumber;

Results :-

```
mysql> select p.checknumber, c.customername, p.amount
    -> from payments p
    -> join customers c
    -> on p.customernumber = c.customernumber;
+-------------+--------------+--------+
| checknumber | customername | amount |
+-------------+--------------+--------+
| CHK001      | ABC Corp     | 300.00 |
| CHK002      | XYZ Ltd      | 500.00 |
| CHK003      | Global Inc   | 150.00 |
+-------------+--------------+--------+
3 rows in set (0.00 sec)
```

11. Count total number of customers

Query :- select count(*) as total_customers from customers;

Results :-

```
mysql> select count(*) as total_customers from customers;
+-----------------+
| total_customers |
+-----------------+
|               3 |
+-----------------+
1 row in set (0.02 sec)
```

12. Count total number of orders

Query :- select count(*) as total_orders from orders;

Results :-

```
mysql> select count(*) as total_orders from orders;
+--------------+
| total_orders |
+--------------+
|            3 |
+--------------+
1 row in set (0.00 sec)
```

13. Find average buyPrice of products

Query :- select avg(buyprice) as avg_price from products;

Results :-

```
mysql> select avg(buyprice) as avg_price from products;
+-----------+
| avg_price |
+-----------+
| 16.000000 |
+-----------+
1 row in set (0.00 sec)
```

14. Find maximum MSRP from products

Query :- select max(msrp) as max_price from products;

Results :-

```
mysql> select max(msrp) as max_price from products;
+-----------+
| max_price |
+-----------+
|     30.00 |
+-----------+
1 row in set (0.02 sec)
```

15. Find total number of employees

Query :- select count(*) as total_employees from employees;

Results :-

```
mysql> select count(*) as total_employees from employees;
+-----------------+
| total_employees |
+-----------------+
|               3 |
+-----------------+
1 row in set (0.00 sec)
```

16. Count customers per country

Query :- select country, count(*) as total_customers
from customers
group by country;

Results :-

```
mysql> select country, count(*) as total_customers
    -> from customers
    -> group by country;
+---------+-----------------+
| country | total_customers |
+---------+-----------------+
| India   |               1 |
| UK      |               1 |
| USA     |               1 |
+---------+-----------------+
3 rows in set (0.02 sec)
```

17. Count orders per status

Query :- select productline, count(*) as total_products
from products
group by productline;

Results :-

```
mysql> select productline, count(*) as total_products
    -> from products
    -> group by productline;
+-------------+----------------+
| productline | total_products |
+-------------+----------------+
| Bikes       |              1 |
| Cars        |              2 |
+-------------+----------------+
2 rows in set (0.00 sec)
```

18. Count products per product line

Query :- select productline, count(*) as total_products
from products
group by productline;

Results :-

```
mysql> select productline, count(*) as total_products
    -> from products
    -> group by productline;
+-------------+----------------+
| productline | total_products |
+-------------+----------------+
| Bikes       |              1 |
| Cars        |              2 |
+-------------+----------------+
2 rows in set (0.00 sec)
```

19. Find average credit limit per country

Query :- select country, avg(creditlimit) as avg_credit
from customers
group by country;

Results :-

```
mysql> select country, avg(creditlimit) as avg_credit
    -> from customers
    -> group by country;
+---------+--------------+
| country | avg_credit   |
+---------+--------------+
| India   | 60000.000000 |
| UK      | 70000.000000 |
| USA     | 50000.000000 |
+---------+--------------+
3 rows in set (0.02 sec)
```

20. Count employees per office

Query :- select officecode, count(*) as total_employees
from employees
group by officecode;

Results :-

```
mysql> select officecode, count(*) as total_employees
    -> from employees
    -> group by officecode;
+------------+-----------------+
| officecode | total_employees |
+------------+-----------------+
| 1          |               2 |
| 2          |               1 |
+------------+-----------------+
2 rows in set (0.02 sec)
```

21. count number of orders per customer

Query :- select customerNumber, count(*) as total_orders
from orders
group by customerNumber;

Results :-

```
mysql> select customerNumber, count(*) as total_orders
    -> from orders
    -> group by customerNumber;
+----------------+--------------+
| customerNumber | total_orders |
+----------------+--------------+
|           2001 |            1 |
|           2002 |            1 |
|           2003 |            1 |
+----------------+--------------+
3 rows in set (0.00 sec)
```

22. find total payment amount per customer

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
    -> from payments
    -> group by customerNumber;
+----------------+---------------+
| customerNumber | total_payment |
+----------------+---------------+
|           2001 |        300.00 |
|           2002 |        500.00 |
|           2003 |        150.00 |
+----------------+---------------+
3 rows in set (0.00 sec)
```

23. list total quantity ordered per product

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
    -> from orderdetails
    -> group by productCode;
+-------------+----------------+
| productCode | total_quantity |
+-------------+----------------+
| P101        |             13 |
| P102        |              7 |
| P201        |              5 |
+-------------+----------------+
3 rows in set (0.00 sec)
```

24. show total revenue per product

Query :- select productCode, sum(quantityOrdered * priceEach) as total_revenue
from orderdetails
group by productCode;

Results :-

```
mysql> select productCode, sum(quantityOrdered * priceEach) as total_revenue
    -> from orderdetails
    -> group by productCode;
+-------------+---------------+
| productCode | total_revenue |
+-------------+---------------+
| P101        |        390.00 |
| P102        |        196.00 |
| P201        |         75.00 |
+-------------+---------------+
3 rows in set (0.01 sec)
```

25. find number of customers handled by each employee

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
    -> from customers
    -> group by salesRepEmployeeNumber;
+------------------------+-----------------+
| salesRepEmployeeNumber | total_customers |
+------------------------+-----------------+
|                   1002 |               2 |
|                   1003 |               1 |
+------------------------+-----------------+
2 rows in set (0.02 sec)
```

26. find customers with more than 5 orders

Query :- select customerNumber, count(*) as total_orders
from orders
group by customerNumber
having count(*) > 5;

Results :-

```
mysql> select customerNumber, count(*) as total_orders
    -> from orders
    -> group by customerNumber
    -> having count(*) > 5;
Empty set (0.02 sec)
```

27. list products with total quantity ordered > 100

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode
having sum(quantityOrdered) > 100;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
    -> from orderdetails
    -> group by productCode
    -> having sum(quantityOrdered) > 100;
Empty set (0.01 sec)
```

28. find employees managing more than 3 customers

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber
having count(*) > 3;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
    -> from customers
    -> group by salesRepEmployeeNumber
    -> having count(*) > 3;
Empty set (0.00 sec)
```

29. show countries with more than 10 customers

Query :- select country, count(*) as total_customers
from customers
group by country
having count(*) > 10;

Results :-

```
mysql> select country, count(*) as total_customers
    -> from customers
    -> group by country
    -> having count(*) > 10;
Empty set (0.00 sec)
```

30. find product lines having more than 5 products

Query :- select productLine, count(*) as total_products
from products
group by productLine
having count(*) > 5;

Results :-

```
mysql> select productLine, count(*) as total_products
    -> from products
    -> group by productLine
    -> having count(*) > 5;
Empty set (0.00 sec)
```

31. list top 5 customers by total payment amount

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber
order by total_payment desc
limit 5;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
    -> from payments
    -> group by customerNumber
    -> order by total_payment desc
    -> limit 5;
+----------------+---------------+
| customerNumber | total_payment |
+----------------+---------------+
|           2002 |        500.00 |
|           2001 |        300.00 |
|           2003 |        150.00 |
+----------------+---------------+
3 rows in set (0.02 sec)
```

32. show products ordered by total sales descending

Query :- select productCode, sum(quantityOrdered * priceEach) as total_sales
from orderdetails
group by productCode
order by total_sales desc;

Results :-

```
mysql> select productCode, sum(quantityOrdered * priceEach) as total_sales
    -> from orderdetails
    -> group by productCode
    -> order by total_sales desc;
+-------------+-------------+
| productCode | total_sales |
+-------------+-------------+
| P101        |      390.00 |
| P102        |      196.00 |
| P201        |       75.00 |
+-------------+-------------+
3 rows in set (0.02 sec)
```

33. list employees sorted by number of customers handled

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber
order by total_customers desc;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
    -> from customers
    -> group by salesRepEmployeeNumber
    -> order by total_customers desc;
+------------------------+-----------------+
| salesRepEmployeeNumber | total_customers |
+------------------------+-----------------+
|                   1002 |               2 |
|                   1003 |               1 |
+------------------------+-----------------+
2 rows in set (0.00 sec)
```

34. show countries sorted by total customers

Query :- select country, count(*) as total_customers
from customers
group by country
order by total_customers desc;

Results :-

```
mysql> select country, count(*) as total_customers
    -> from customers
    -> group by country
    -> order by total_customers desc;
+---------+-----------------+
| country | total_customers |
+---------+-----------------+
| USA     |               1 |
| UK      |               1 |
| India   |               1 |
+---------+-----------------+
3 rows in set (0.00 sec)
```

35. display top 10 most ordered products

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode
order by total_quantity desc
limit 10;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
    -> from orderdetails
    -> group by productCode
    -> order by total_quantity desc
    -> limit 10;
+-------------+----------------+
| productCode | total_quantity |
+-------------+----------------+
| P101        |             13 |
| P102        |              7 |
| P201        |              5 |
+-------------+----------------+
3 rows in set (0.00 sec)
```

36. get customer name, order number, and product name

Query :- select c.customerName, o.orderNumber, p.productName
from orders o
join customers c on o.customerNumber = c.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select c.customerName, o.orderNumber, p.productName
    -> from orders o
    -> join customers c on o.customerNumber = c.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> join products p on od.productCode = p.productCode;
+--------------+-------------+--------------+
| customerName | orderNumber | productName  |
+--------------+-------------+--------------+
| ABC Corp     |        3001 | Toyota Camry |
| Global Inc   |        3003 | Toyota Camry |
| XYZ Ltd      |        3002 | Honda Civic  |
| ABC Corp     |        3001 | Yamaha Bike  |
+--------------+-------------+--------------+
4 rows in set (0.02 sec)
```

37. show employee name, customer name, and total payments

Query :- select e.firstName, e.lastName, c.customerName, sum(p.amount) as total_payment
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join payments p on c.customerNumber = p.customerNumber
group by e.employeeNumber, c.customerNumber;

Results :-

```
mysql> select e.firstName, e.lastName, c.customerName, sum(p.amount) as total_payment
    -> from employees e
    -> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> join payments p on c.customerNumber = p.customerNumber
    -> group by e.employeeNumber, c.customerNumber;
+-----------+----------+--------------+---------------+
| firstName | lastName | customerName | total_payment |
+-----------+----------+--------------+---------------+
| Jane      | Doe      | ABC Corp     |        300.00 |
| Jane      | Doe      | Global Inc   |        150.00 |
| Mike      | Brown    | XYZ Ltd      |        500.00 |
+-----------+----------+--------------+---------------+
3 rows in set (0.00 sec)
```

38. list order details with product line and quantity ordered

Query :- select od.orderNumber, p.productLine, od.quantityOrdered
from orderdetails od
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select od.orderNumber, p.productLine, od.quantityOrdered
    -> from orderdetails od
    -> join products p on od.productCode = p.productCode;
+-------------+-------------+-----------------+
| orderNumber | productLine | quantityOrdered |
+-------------+-------------+-----------------+
|        3001 | Cars        |              10 |
|        3001 | Bikes       |               5 |
|        3002 | Cars        |               7 |
|        3003 | Cars        |               3 |
+-------------+-------------+-----------------+
4 rows in set (0.00 sec)
```

39. display office city with employee count

Query :- select o.city, count(e.employeeNumber) as total_employees
from offices o
join employees e on o.officeCode = e.officeCode
group by o.city;

Results :-

```
mysql> select o.city, count(e.employeeNumber) as total_employees
    -> from offices o
    -> join employees e on o.officeCode = e.officeCode
    -> group by o.city;
+----------+-----------------+
| city     | total_employees |
+----------+-----------------+
| London   |               1 |
| New York |               2 |
+----------+-----------------+
2 rows in set (0.00 sec)
```

40. show product vendor with total quantity sold

Query :- select p.productVendor, sum(od.quantityOrdered) as total_quantity
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productVendor;

Results :-

```
mysql> select p.productVendor, sum(od.quantityOrdered) as total_quantity
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productVendor;
+---------------+----------------+
| productVendor | total_quantity |
+---------------+----------------+
| Honda         |              7 |
| Toyota        |             13 |
| Yamaha        |              5 |
+---------------+----------------+
3 rows in set (0.00 sec)
```

41. find average order value per customer

Query :- select o.customerNumber, avg(od.quantityOrdered * od.priceEach) as avg_order_value
from orders o
join orderdetails od on o.orderNumber = od.orderNumber
group by o.customerNumber;

Results :-

```
mysql> select o.customerNumber, avg(od.quantityOrdered * od.priceEach) as avg_order_value
    -> from orders o
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by o.customerNumber;
+----------------+-----------------+
| customerNumber | avg_order_value |
+----------------+-----------------+
|           2001 |      187.500000 |
|           2002 |      196.000000 |
|           2003 |       90.000000 |
+----------------+-----------------+
3 rows in set (0.00 sec)
```

42. show total revenue generated per country

Query :- select c.country, sum(od.quantityOrdered * od.priceEach) as total_revenue
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.country;

Results :-

```
mysql> select c.country, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by c.country;
+---------+---------------+
| country | total_revenue |
+---------+---------------+
| India   |         90.00 |
| UK      |        196.00 |
| USA     |        375.00 |
+---------+---------------+
3 rows in set (0.00 sec)
```

43. find customers whose total payments exceed 50000

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber
having sum(amount) > 50000;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
    -> from payments
    -> group by customerNumber
    -> having sum(amount) > 50000;
Empty set (0.00 sec)
```

44. list products where average order quantity > 50

Query :- select productCode, avg(quantityOrdered) as avg_qty
from orderdetails
group by productCode
having avg(quantityOrdered) > 50;

Results :-

```
mysql> select productCode, avg(quantityOrdered) as avg_qty
    -> from orderdetails
    -> group by productCode
    -> having avg(quantityOrdered) > 50;
Empty set (0.00 sec)
```

45. find top 3 product lines by total revenue

Query :- select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productLine
order by total_revenue desc
limit 3;

Results :-

```
mysql> select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productLine
    -> order by total_revenue desc
    -> limit 3;
+-------------+---------------+
| productLine | total_revenue |
+-------------+---------------+
| Cars        |        586.00 |
| Bikes       |         75.00 |
+-------------+---------------+
2 rows in set (0.00 sec)
```

46. show customers with credit limit > average credit limit

Query :- select *
from customers
where creditLimit > (select avg(creditLimit) from customers);

Results :-

```
mysql> select *
    -> from customers
    -> where creditLimit > (select avg(creditLimit) from customers);
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
| customerNumber | customerName | contactLastName | contactFirstName | phone      | addressLine1 | addressLine2 | city   | state | postalCode | country | salesRepEmployeeNumber | creditLimit |
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
|           2002 | XYZ Ltd      | Kim             | Pat              | 2222222222 | Street 2     | NULL         | London | NULL  | NW1        | UK      |                   1003 |    70000.00 |
+----------------+--------------+-----------------+------------------+------------+--------------+--------------+--------+-------+------------+---------+------------------------+-------------+
1 row in set (0.01 sec)
```

47. find orders where total amount > 10000

Query :- select orderNumber, sum(quantityOrdered * priceEach) as total_amount
from orderdetails
group by orderNumber
having sum(quantityOrdered * priceEach) > 10000;

Results :-

```
mysql>
mysql> select orderNumber, sum(quantityOrdered * priceEach) as total_amount
    -> from orderdetails
    -> group by orderNumber
    -> having sum(quantityOrdered * priceEach) > 10000;
Empty set (0.00 sec)
```

48. list employees working in offices with more than 5 employees

Query :- select *
from employees
where officeCode in (
select officeCode
from employees
group by officeCode
having count(*) > 5
);

Results :-

```
mysql> select *
    -> from employees
    -> where officeCode in (
    -> select officeCode
    -> from employees
    -> group by officeCode
    -> having count(*) > 5
    -> );
Empty set (0.00 sec)
```

49. find products with stock less than average stock

Query :- select *
from products
where quantityInStock < (select avg(quantityInStock) from products);

Results :-

```
mysql> select *
    -> from products
    -> where quantityInStock < (select avg(quantityInStock) from products);
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
| productCode | productName  | productLine | productScale | productVendor | productDescription | quantityInStock | buyPrice | MSRP  |
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
| P101        | Toyota Camry | Cars        | 1:18         | Toyota        | Sedan model        |             100 |    20.00 | 30.00 |
+-------------+--------------+-------------+--------------+---------------+--------------------+-----------------+----------+-------+
1 row in set (0.01 sec)
```

50. show customers who made more than 3 payments

Query :- select customerNumber, count(*) as total_payments
from payments
group by customerNumber
having count(*) > 3;

Results :-

```
mysql> select customerNumber, count(*) as total_payments
    -> from payments
    -> group by customerNumber
    -> having count(*) > 3;
Empty set (0.00 sec)
```

51. find total revenue generated by each customer

Query :- select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName
order by total_revenue desc;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by c.customerName
    -> order by total_revenue desc;
+--------------+---------------+
| customerName | total_revenue |
+--------------+---------------+
| ABC Corp     |        375.00 |
| XYZ Ltd      |        196.00 |
| Global Inc   |         90.00 |
+--------------+---------------+
3 rows in set (0.00 sec)
```

52. list customers who have placed orders but made no payments

Query :- select distinct c.customerName
from customers c
join orders o on c.customerNumber = o.customerNumber
left join payments p on c.customerNumber = p.customerNumber
where p.customerNumber is null;

Results :-

```
mysql> select distinct c.customerName
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> left join payments p on c.customerNumber = p.customerNumber
    -> where p.customerNumber is null;
Empty set (0.01 sec)
```

53. find total number of products ordered by each customer

Query :- select c.customerName, sum(od.quantityOrdered) as total_products,
sum(od.quantityOrdered * od.priceEach) as total_amount
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered) as total_products,
    -> sum(od.quantityOrdered * od.priceEach) as total_amount
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by c.customerName;
+--------------+----------------+--------------+
| customerName | total_products | total_amount |
+--------------+----------------+--------------+
| ABC Corp     |             15 |       375.00 |
| Global Inc   |              3 |        90.00 |
| XYZ Ltd      |              7 |       196.00 |
+--------------+----------------+--------------+
3 rows in set (0.00 sec)
```

54. show product-wise revenue and total quantity sold

Query :- select p.productName,
sum(od.quantityOrdered) as total_qty,
sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productName
order by total_revenue desc;

Results :-

```
mysql> select p.productName,
    -> sum(od.quantityOrdered) as total_qty,
    -> sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productName
    -> order by total_revenue desc;
+--------------+-----------+---------------+
| productName  | total_qty | total_revenue |
+--------------+-----------+---------------+
| Toyota Camry |        13 |        390.00 |
| Honda Civic  |         7 |        196.00 |
| Yamaha Bike  |         5 |         75.00 |
+--------------+-----------+---------------+
3 rows in set (0.00 sec)
```

55. find employees along with total sales generated

Query :- select e.firstName, e.lastName,
sum(od.quantityOrdered * od.priceEach) as total_sales
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by e.employeeNumber;

Results :-

```
mysql> select e.firstName, e.lastName,
    -> sum(od.quantityOrdered * od.priceEach) as total_sales
    -> from employees e
    -> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by e.employeeNumber;
+-----------+----------+-------------+
| firstName | lastName | total_sales |
+-----------+----------+-------------+
| Jane      | Doe      |      465.00 |
| Mike      | Brown    |      196.00 |
+-----------+----------+-------------+
2 rows in set (0.00 sec)
```

56. find customers whose total purchase amount is greater than average

Query :-

Results :-

57. list products ordered in more than 2 different orders

Query :- select productCode, count(distinct orderNumber) as total_orders
from orderdetails
group by productCode
having count(distinct orderNumber) > 2;

Results :-

```
mysql> select productCode, count(distinct orderNumber) as total_orders
    -> from orderdetails
    -> group by productCode
    -> having count(distinct orderNumber) > 2;
Empty set (0.00 sec)
```

58. find customers who ordered more than 3 different products

Query :- select o.customerNumber, count(distinct od.productCode) as total_products
from orders o
join orderdetails od on o.orderNumber = od.orderNumber
group by o.customerNumber
having count(distinct od.productCode) > 3;

Results :-

```
mysql> select o.customerNumber, count(distinct od.productCode) as total_products
    -> from orders o
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by o.customerNumber
    -> having count(distinct od.productCode) > 3;
Empty set (0.02 sec)
```

59. show product lines where total revenue exceeds 500

Query :- select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productLine
having total_revenue > 500;

Results :-

```
mysql> select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productLine
    -> having total_revenue > 500;
+-------------+---------------+
| productLine | total_revenue |
+-------------+---------------+
| Cars        |        586.00 |
+-------------+---------------+
1 row in set (0.00 sec)
```

60. find countries where average credit limit > 60000

Query :- select country, avg(creditLimit) as avg_credit
from customers
group by country
having avg_credit > 60000;

Results :-

```
mysql> select country, avg(creditLimit) as avg_credit
    -> from customers
    -> group by country
    -> having avg_credit > 60000;
+---------+--------------+
| country | avg_credit   |
+---------+--------------+
| UK      | 70000.000000 |
+---------+--------------+
1 row in set (0.01 sec)
```

61. show customer name, order number, product name, and quantity ordered for all orders

Query :- select c.customerName, o.orderNumber, p.productName, od.quantityOrdered
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select c.customerName, o.orderNumber, p.productName, od.quantityOrdered
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> join products p on od.productCode = p.productCode;
+--------------+-------------+--------------+-----------------+
| customerName | orderNumber | productName  | quantityOrdered |
+--------------+-------------+--------------+-----------------+
| ABC Corp     |        3001 | Toyota Camry |              10 |
| ABC Corp     |        3001 | Yamaha Bike  |               5 |
| XYZ Ltd      |        3002 | Honda Civic  |               7 |
| Global Inc   |        3003 | Toyota Camry |               3 |
+--------------+-------------+--------------+-----------------+
4 rows in set (0.00 sec)
```

62. find employee name, office city, and number of customers handled

Query :- select e.firstName, e.lastName, o.city, count(c.customerNumber) as total_customers
from employees e
join offices o on e.officeCode = o.officeCode
left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
group by e.employeeNumber;

Results :-

```
mysql> select e.firstName, e.lastName, o.city, count(c.customerNumber) as total_customers
    -> from employees e
    -> join offices o on e.officeCode = o.officeCode
    -> left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> group by e.employeeNumber;
+-----------+----------+----------+-----------------+
| firstName | lastName | city     | total_customers |
+-----------+----------+----------+-----------------+
| John      | Smith    | New York |               0 |
| Jane      | Doe      | New York |               2 |
| Mike      | Brown    | London   |               1 |
+-----------+----------+----------+-----------------+
3 rows in set (0.00 sec)
```

63. display product line, product name, and total revenue generated

Query :- select p.productLine, p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productCode;

Results :-

```
mysql> select p.productLine, p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productCode;
+-------------+--------------+---------------+
| productLine | productName  | total_revenue |
+-------------+--------------+---------------+
| Cars        | Toyota Camry |        390.00 |
| Cars        | Honda Civic  |        196.00 |
| Bikes       | Yamaha Bike  |         75.00 |
+-------------+--------------+---------------+
3 rows in set (0.00 sec)
```

64. show order number, customer name, and total order value

Query :- select o.orderNumber, c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
from orders o
join customers c on o.customerNumber = c.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by o.orderNumber;

Results :-

```
mysql> select o.orderNumber, c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
    -> from orders o
    -> join customers c on o.customerNumber = c.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by o.orderNumber;
+-------------+--------------+-------------+
| orderNumber | customerName | total_value |
+-------------+--------------+-------------+
|        3001 | ABC Corp     |      375.00 |
|        3002 | XYZ Ltd      |      196.00 |
|        3003 | Global Inc   |       90.00 |
+-------------+--------------+-------------+
3 rows in set (0.00 sec)
```

65. list employees whose customers have placed more than 5 orders in total

Query :- select e.employeeNumber, e.firstName, e.lastName
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join orders o on c.customerNumber = o.customerNumber
group by e.employeeNumber
having count(o.orderNumber) > 5;

Results :-

```
mysql> select e.employeeNumber, e.firstName, e.lastName
    -> from employees e
    -> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> join orders o on c.customerNumber = o.customerNumber
    -> group by e.employeeNumber
    -> having count(o.orderNumber) > 5;
Empty set (0.00 sec)
```

66. find customers whose total payments are less than their total order value
Query :-
Results :-

67. list products where total quantity sold is less than average quantity sold across all products
Query :-
Results :-

68. find customers who have not placed any orders but have made payments
Query :- select c.customerName
from customers c
left join orders o on c.customerNumber = o.customerNumber
join payments p on c.customerNumber = p.customerNumber
where o.orderNumber is null;
Results :-

```
mysql> select c.customerName
    -> from customers c
    -> left join orders o on c.customerNumber = o.customerNumber
    -> join payments p on c.customerNumber = p.customerNumber
    -> where o.orderNumber is null;
Empty set (0.00 sec)
```

69. show employees who have not been assigned any customers
Query :- select *
from employees e
left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
where c.customerNumber is null;
Results :-

```
mysql> select *
    -> from employees e
    -> left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> where c.customerNumber is null;
+----------------+----------+-----------+-----------+------------------+------------+-----------+----------+----------------+--------------+-----------------+------------------+-------+--------------+--------------+------+-------+------------+---------+------------------------+-------------+
| employeeNumber | lastName | firstName | extension | email            | officeCode | reportsTo | jobTitle | customerNumber | customerName | contactLastName | contactFirstName | phone | addressLine1 | addressLine2 | city | state | postalCode | country | salesRepEmployeeNumber | creditLimit |
+----------------+----------+-----------+-----------+------------------+------------+-----------+----------+----------------+--------------+-----------------+------------------+-------+--------------+--------------+------+-------+------------+---------+------------------------+-------------+
|           1001 | Smith    | John      | x101      | john@company.com | 1          |      NULL | Manager  |           NULL | NULL         | NULL            | NULL             | NULL  | NULL         | NULL         | NULL | NULL  | NULL       | NULL    |                   NULL |        NULL |
+----------------+----------+-----------+-----------+------------------+------------+-----------+----------+----------------+--------------+-----------------+------------------+-------+--------------+--------------+------+-------+------------+---------+------------------------+-------------+
1 row in set (0.00 sec)

mysql> |
```

70. find orders that contain more than 1 product (multi-line orders)

Query :- select orderNumber, count(productCode) as total_products
from orderdetails
group by orderNumber
having count(productCode) > 1;

Results :-

```
mysql> select orderNumber, count(productCode) as total_products
    -> from orderdetails
    -> group by orderNumber
    -> having count(productCode) > 1;
+-------------+----------------+
| orderNumber | total_products |
+-------------+----------------+
|        3001 |              2 |
+-------------+----------------+
1 row in set (0.00 sec)
```

71. find top 3 customers based on total order value

Query :- select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName
order by total_value desc
limit 3;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
    -> from customers c
    -> join orders o on c.customerNumber = o.customerNumber
    -> join orderdetails od on o.orderNumber = od.orderNumber
    -> group by c.customerName
    -> order by total_value desc
    -> limit 3;
+--------------+-------------+
| customerName | total_value |
+--------------+-------------+
| ABC Corp     |      375.00 |
| XYZ Ltd      |      196.00 |
| Global Inc   |       90.00 |
+--------------+-------------+
3 rows in set (0.00 sec)
```

72. show top 5 products based on total revenue

Query :- select p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productName
order by total_revenue desc
limit 5;

Results :-

```
mysql> select p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
    -> from products p
    -> join orderdetails od on p.productCode = od.productCode
    -> group by p.productName
    -> order by total_revenue desc
    -> limit 5;
+--------------+---------------+
| productName  | total_revenue |
+--------------+---------------+
| Toyota Camry |        390.00 |
| Honda Civic  |        196.00 |
| Yamaha Bike  |         75.00 |
+--------------+---------------+
3 rows in set (0.00 sec)
```

73. list top 2 employees based on sales generated

Query :-

Results :-

74. find top 3 countries by number of customers

Query :- select country, count(*) as total_customers
from customers
group by country
order by total_customers desc
limit 3;

Results :-

```
mysql> select country, count(*) as total_customers
    -> from customers
    -> group by country
    -> order by total_customers desc
    -> limit 3;
+---------+-----------------+
| country | total_customers |
+---------+-----------------+
| USA     |               1 |
| UK      |               1 |
| India   |               1 |
+---------+-----------------+
3 rows in set (0.00 sec)
```

75. show top 5 most expensive products based on msrp

Query :- select productName, msrp
from products
order by msrp desc
limit 5;

Results :-

```
mysql> select productName, msrp
    -> from products
    -> order by msrp desc
    -> limit 5;
+--------------+-------+
| productName  | msrp  |
+--------------+-------+
| Toyota Camry | 30.00 |
| Honda Civic  | 28.00 |
| Yamaha Bike  | 15.00 |
+--------------+-------+
3 rows in set (0.00 sec)
```

76. find average order value per customer and list only those above overall average
Query :-
Results :-

77. show product-wise average price sold vs msrp difference
Query :-
Results :-

78. find percentage contribution of each product to total revenue
Query :-
Results :-

79. calculate average quantity ordered per order for each customer
Query :-
Results :-

80. find customers whose largest single order value exceeds 10000
Query :-
Results :-